

KIRANA GPT

A loan officer uploads 5 photos.

KiranaGPT approves the loan in 60 seconds — no field visit, no paperwork.

Multi-Agent AI Credit Intelligence for India's 13M Kirana Stores

13M+

Kirana stores in India

₹40L Cr

Addressable credit market

< 60s

End-to-end AI decision

6

AI agents (Gemini-powered)

Zero

Field visits required

India's 13 Million Stores Can't Get a Loan

13M+

Kirana stores
across India

₹0

Formal financial
records kept

2-4 Wks

Manual field
survey timeline

₹5,000

Cost per field
visit — paid by NBFC

✗ Today's Reality

- ✗ Field officer physically visits every store
- ✗ Owners coach answers — easily gamed
- ✗ No reproducible or auditable scoring
- ✗ ₹5,000 cost + 2-4 weeks per decision
- ✗ Result: 90% of creditworthy stores rejected

✓ What the Market Needs

- ✓ Remote underwriting — zero field visits
- ✓ Objective, fraud-resistant AI scoring
- ✓ Sub-minute decision at near-zero cost
- ✓ Scales to every PIN code in India
- ✓ Explainable AI for RBI compliance

KiranaGPT: 5 Photos → Loan Decision in 60 Seconds



What the AI Actually Tells You — Example Output

Business Advisor Agent

Business Health Score

72/100

Top Fix: Stock Maggi, Parle-G, Kurkure — adds
₹10K/month

Opportunity: Add recharge counter — +20%
footfall

Loan Advisor Agent

Recommended Loan

₹2,50,000

Based on ₹1.5L monthly revenue, 8 years
tenure, Tier-1 location. XGBoost score 712. Risk
band: LOW. EMI: ₹24,583/month.

Voice Agent (Hindi)

Owner asks in Hindi:

"Mera business kaise badhega?"

AI responds in Hindi (voice):

*"Maggi aur Parle-G stock karein — isse aapka
revenue 18% badh sakta hai."*

Full-Stack AI — Every Layer Purpose-Built

AI & Computer Vision

YOLOv8n (Ultralytics)

Object detection across 5 images

Google Gemini 1.5 Flash

LLM reasoning — all 6 agents (free)

Web Speech API

Speech-to-text (STT)

Browser SpeechSynthesis API

Text-to-speech (Hindi/EN)

XGBoost / sklearn

ML credit scoring (300–900)

Backend & Data

FastAPI + Uvicorn

API server & routing

OpenCV + CLAHE

Image preprocessing

OSM Overpass API

Real-time geo intelligence

Markdown + HTML Reports

AI-generated, downloadable

Pydantic

Env config, validation, schemas

Frontend & Database

Next.js 14 (App Router)

Frontend framework

TypeScript + Tailwind CSS

UI language & styling

Supabase / PostgreSQL

Database + audit trail

python-dotenv + asyncio

Env management + async agents

React hooks + API

State management & calls

6-Agent Autonomous AI System

INPUT → 5 shop images + GPS coordinates + optional details (rent · size · years)

ORCHESTRATOR AGENT (Gemini 1.5 Flash)

Manages state · calls all agents · handles retries · synthesises final response

Vision Agent

YOLOv8n +
Gemini insight

Geo Intel Agent

OSM live
market data

Credit Risk Agent

XGBoost +
LLM narrative

Business Advisor ★

Gemini strategy
& SKU advice

Loan Advisor ★

Lender-grade
credit memo

Report Agent

PDF · HTML · Markdown

Voice Agent ★

Hindi + English STT/TTS

OUTPUT

Revenue range · Fraud flags · Business advice · Lender loan memo · PDF report · Hindi voice summary · Supabase audit trail

How We Build It — 9 Steps, Fully Automated

Core Pipeline

①	ImageLoader	Load & preprocess 5 images · gamma correction · CLAHE denoising
②	YOLODetector	YOLOv8n across all images · detections sorted by confidence
③	ShelfAnalyzer	SDI metrics · zone analysis · uniformity · depth scoring
④	InventoryEstimator	Map YOLO detections to INR value bands: staples / FMCG / high-margin
⑤	VisualProcessor	6-component weighted score → visual_score (0–1)
⑥	GeoFeatureExtractor	Live OSM API · population rings · POI · competition · road type
⑦	FraudDetector	12 cross-signal rules: visual / geo / cross-signal fraud detection
⑧	FusionModel	$\text{composite} = 0.40 \times \text{visual} + 0.35 \times \text{geo} - 0.25 \times \text{fraud} \rightarrow \text{APPROVE/REVIEW/REJECT}$
⑨	LLM Agents	Gemini Orchestrator calls all agents in parallel → advice, memo, PDF, voice

Gemini Agent Layer

Business Advisor	Inventory advice, revenue strategy, health score
Loan Advisor	Lender-grade credit memo + EMI sizing
Credit Risk	XGBoost score + LLM risk narrative
Voice Agent	Hindi/English STT → Gemini → TTS response

Fraud Detection — 12 Rules

CRITICAL	Identical images uploaded across slots
HIGH	shelf_occupancy < 0.10 or image blurry
HIGH	Shop size > 800 sqft but < 15 items detected
MEDIUM	competitor_count > 15 within 300m
MEDIUM	revenue > ₹5L but shelf_occupancy < 0.20

Business Advisor Agent — Gemini at Its Best

Agent Inputs

- YOLOv8 inventory value & category mix
- SDI shelf occupancy score
- Geo market score + competitor count
- City tier & population density
- XGBoost credit score (300–900)
- Fraud flags + composite risk score
- Years in operation + monthly rent

Gemini Business Advisor — Full Example Output

72Health Score
/ 100**+18%**Revenue
Uplift**LOW**Risk Band
(AI + ML)

⚠️ Top Problems (AI-identified)

FMCG variety too low — missing Maggi, Parle-G, Kurkure SKUs
High competition — 4 stores within 300m radius

★ Top Opportunities

Add mobile recharge counter — increases daily footfall by ~20%
Stock premium biscuits (Dark Fantasy, Oreo) — 40% gross margin

Revenue Growth Strategy (AI-generated)

Stock Maggi, Parle-G, Kurkure at the billing counter — these 3 SKUs alone add ₹8K–₹12K/month. Partner with a local dairy for morning delivery to extend the store's trading window beyond evening hours.

Fully Autonomous — No Human in the Loop

Live Agent Execution Log

✓ Image Loader	0.4s
5 images loaded, 2 low-light enhanced	
✓ Vision Agent	1.2s
84 items, SDI=0.71, visual_score=0.68	
✓ Geo Agent	2.1s
Mumbai T1, 4 rivals in 300m, geo=0.74	
✓ Credit Risk	0.8s
XGBoost score=712, risk_band=LOW	
✓ Fraud Detector	0.3s
1 flag: VISUAL_LOW_PRODUCTS (medium)	
✓ Business Advisor	3.2s
Health=72, +18% strategy, 3 SKUs	
✓ Loan Advisor	2.8s
₹2,50,000 rec. + lender memo written	
✓ Report Agent	1.5s
PDF + HTML report compiled	
✓ Voice Agent	4.1s
Hindi audio synthesised (48s clip)	

Total: 16.6s end-to-end

Loan Advisor — Sample Output

₹2,50,000

Recommended Loan

12 months

Tenure

₹24,583

Monthly EMI

AI Reasoning (lender-style):

Monthly revenue ₹1.2L-₹1.8L → 4× cap = ₹6L. XGBoost 712 + 1 MEDIUM flag
→ 0.82× multiplier → ₹2,50,000 recommended. 8 years tenure + Mumbai T1
are strong positive signals.



Voice Agent — Hindi + English

Owner asks (Hindi voice input):

"Mujhe business improve karne ke liye kya karna chahiye?"

AI responds in Hindi (synthesised voice):

"Maggi, Parle-G stock karein — revenue 18% badh sakta hai."

A Credit Bureau for India's Physical Economy

13M

Kirana stores
addressable — immediately

₹5,000 → ₹0

Field visit cost
reduced per assessment

< 60s

Photo upload to
full AI credit decision

3 Groups

Who directly
benefit from KiranaGPT

Who Benefits



Kirana Store Owners

Access formal credit for the first time in their lives. Get AI business advice to grow revenue. No paperwork, no branch visits, no intimidation. Results in 60 seconds, not weeks.



NBFCs & Banks

Cut field visit costs from ₹5,000 to near-zero. Scale loan disbursements to underserved geographies overnight. AI-explainable decisions satisfy RBI compliance requirements.



Indian Economy

Formalise ₹40L Cr informal retail economy. Advance financial inclusion for India's 13M unbanked micro-retailers. Aligned with Digital India, Jan Dhan Yojana, and PM Mudra goals.

Roadmap to Full Production Deployment

Phase 1

Live APIs

Enhance live OSM Overpass API with WorldPop + Google Places for richer population rings. Add Redis caching layer for sub-second geo responses and real-time competition feed.

Phase 2

Fine-tune ML Models

Partner with NBFC to collect real kirana loan outcomes. Re-train XGBoost on actual default data to replace synthetic training set. Backtest and validate composite scores against historical approval rates.

Phase 3

WhatsApp Bot

Loan officers submit via WhatsApp — photo + GPS pin. KiranaGPT responds with full analysis in 60s. Zero app install. Full vernacular language support (Hindi, Tamil, Telugu).

Phase 4

NBFC Deployment

White-label API for NBFCs (Poonawalla, Aye Finance, Stashfin). LOS dashboard integration. RBI-compliant audit trail + AI explainability reports for every decision.

Generative AI ✓

Agentic AI ✓

Social Impact ✓

Team: Aryan Raj · Shashank Pandey · Anushka Dixit · HackArena 2.0 · Gemini + YOLOv8 + OSM · Generative & Agentic AI